Code No: 117EE **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, October/November - 2020
LINUX PROGRAMMING
(Computer Science and Engineering)

Time: 2 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
- - -

1.a) Write a shell scritp to find the factorial of a number.
b) List and explain various Process utilities available in Linux. [7+8]

2.a) How to retrieve the status information of a file? Give the syntax of the system call and illustrate with an example program.
b) Explain lseek( ) functions with an example program. [7+8]

3. Make a comparison between wait( ) and waitpid( ) system calls and explain various macros associated with these system calls. [15]

4.a) How are pipes differ from named pipes?
b) Explain the implementation of IPC using unnamed pipes. [5+10]

5. How to implement IPC between two processes using Shared memory concept? Explain with an example client/server program using shared memory. [15]

6.a) Write a awk script illustrating various string and mathematical functions.
b) Give the syntax of various control structures available in bash shell and explain them briefly. [7+8]

7. List and explain various directory management functions/system calls with clear syntax and example program. [15]

8.a) What is an orphan process? List out the similarities and dissimilarities between a parent process and its child process.
b) Illustrate signal( ) system call with an example program. [7+8]

--ooOoo--

ITUH USED 07-11-2020PM PAP